

EXPERIMENT 2

EXPLORATORY DATA ANALYSIS (EDA) – DATA IMPORT AND EXPORT

AIM:

To import data from various sources such as CSV, Excel, SQL, and Web, perform basic inspection using Pandas, and export the processed data into different file formats.

ALGORITHM 1 :

1. Import Pandas library
2. Load CSV file using read_csv()
3. Display first few records
4. Check data structure
5. Analyze imported data

PROGRAM 1 :

```
In [1]: import pandas as pd
df=pd.read_csv("diabetes.csv")
print(df)
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	\
0	6	148	72	35	0	33.6	
1	1	85	66	29	0	26.6	
2	8	183	64	0	0	23.3	
3	1	89	66	23	94	28.1	
4	0	137	40	35	168	43.1	
..	...	...	...	...	...	...	
763	10	101	76	48	180	32.9	
764	2	122	70	27	0	36.8	
765	5	121	72	23	112	26.2	
766	1	126	60	0	0	30.1	
767	1	93	70	31	0	30.4	

	DiabetesPedigreeFunction	Age	Outcome
0	0.627	50	1
1	0.351	31	0
2	0.672	32	1
3	0.167	21	0
4	2.288	33	1
..	...	...	...
763	0.171	63	0
764	0.340	27	0
765	0.245	30	0
766	0.349	47	1
767	0.315	23	0

[768 rows x 9 columns]

PROGRAM 2 :

```
In [2]: print(df.head())
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI \
0	6	148	72	35	0	33.6
1	1	85	66	29	0	26.6
2	8	183	64	0	0	23.3
3	1	89	66	23	94	28.1
4	0	137	40	35	168	43.1

	DiabetesPedigreeFunction	Age	Outcome
0	0.627	50	1
1	0.351	31	0
2	0.672	32	1
3	0.167	21	0
4	2.288	33	1

PROGRAM 3:

```
In [3]: print(df.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 768 entries, 0 to 767
Data columns (total 9 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Pregnancies                          768 non-null   int64
1   Glucose                             768 non-null   int64
2   BloodPressure                       768 non-null   int64
3   SkinThickness                      768 non-null   int64
4   Insulin                            768 non-null   int64
5   BMI                                768 non-null   float64
6   DiabetesPedigreeFunction            768 non-null   float64
7   Age                                768 non-null   int64
8   Outcome                            768 non-null   int64
dtypes: float64(2), int64(7)
memory usage: 54.1 KB
None
```

PROGRAM 4:

```
In [4]: print(df.describe())
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin \
count	768.000000	768.000000	768.000000	768.000000	768.000000
mean	3.845052	120.894531	69.105469	20.536458	79.799479
std	3.369578	31.972618	19.355807	15.952218	115.244002
min	0.000000	0.000000	0.000000	0.000000	0.000000
25%	1.000000	99.000000	62.000000	0.000000	0.000000
50%	3.000000	117.000000	72.000000	23.000000	30.500000
75%	6.000000	140.250000	80.000000	32.000000	127.250000
max	17.000000	199.000000	122.000000	99.000000	846.000000

	BMI	DiabetesPedigreeFunction	Age	Outcome
count	768.000000	768.000000	768.000000	768.000000
mean	31.992578	0.471876	33.240885	0.348958
std	7.884160	0.331329	11.760232	0.476951
min	0.000000	0.078000	21.000000	0.000000
25%	27.300000	0.243750	24.000000	0.000000
50%	32.000000	0.372500	29.000000	0.000000
75%	36.600000	0.626250	41.000000	1.000000
max	67.100000	2.420000	81.000000	1.000000

ALGORITHM 2:

Importing Excel File Step 1

1. Create Excel file
2. Use read_excel()
3. Display records
4. Verify imported data

PROGRAM 5:

```
In [9]: cf = pd.read_excel("diabetesex.xlsx")
print(cf)
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI \
0	6	148	72	35	0	33.6
1	1	85	66	29	0	26.6
2	8	183	64	0	0	23.3
3	1	89	66	23	94	28.1
4	0	137	40	35	168	43.1
..	...	...	...	...	...	...
763	10	101	76	48	180	32.9
764	2	122	70	27	0	36.8
765	5	121	72	23	112	26.2
766	1	126	60	0	0	30.1
767	1	93	70	31	0	30.4

	DiabetesPedigreeFunction	Age	Outcome
0	0.627	50	1
1	0.351	31	0
2	0.672	32	1
3	0.167	21	0
4	2.288	33	1
..	...	...	...
763	0.171	63	0
764	0.340	27	0
765	0.245	30	0
766	0.349	47	1
767	0.315	23	0

[768 rows x 9 columns]

ALGORITHM 3 :

Importing SQL Data

1. Create SQLite Database
2. Insert records
3. Retrieve records using SQL Query

PROGRAM 6:

```
In [11]: import sqlite3
import pandas as pd
conn = sqlite3.connect("diabetes.db")

query = "SELECT * FROM diabetes"

df = pd.read_sql(query, conn)

print(df)

conn.close()
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	\
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3	1	89	66	23	94	28.1	
4	0	137	40	35	168	43.1	
..	...	...	...	...	...	...	
763	10	101	76	48	180	32.9	
764	2	122	70	27	0	36.8	
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	DiabetesPedigreeFunction	Age	Outcome
0	0.627	50	1
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2	0.672	32	1
3	0.167	21	0
4	2.288	33	1
..	...	...	...
763	0.171	63	0
764	0.340	27	0
765	0.245	30	0
766	0.349	47	1
767	0.315	23	0

[768 rows x 9 columns]

ALGORITHM 4:

Data from Web

1. Specify URL
2. Load data using read_csv()
3. Display imported records

PROGRAM 7:

```
In [13]: import pandas as pd

url = "diabetes.csv"

df = pd.read_csv(url)

print(df.head())
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	\
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2	8	183	64	0	0	23.3	
3	1	89	66	23	94	28.1	
4	0	137	40	35	168	43.1	

	DiabetesPedigreeFunction	Age	Outcome
0	0.627	50	1
1	0.351	31	0
2	0.672	32	1
3	0.167	21	0
4	2.288	33	1

PROGRAM 8:

```
In [15]: import pandas as pd
df = pd.read_csv("diabetes.csv")
df.to_csv("diabetes_output.csv", index=False)
print("CSV File Exported")
```

CSV File Exported

PROGRAM 9:

```
In [16]: import pandas as pd
df = pd.read_csv("diabetes.csv")
df.to_excel("diabetes_output.xlsx", index=False)
print("Excel File Exported")
```

Excel File Exported

PROGRAM 10:

```
In [17]: import pandas as pd
df = pd.read_csv("diabetes.csv")
df.to_json("diabetes_output.json", orient="records")
print("JSON File Exported")
```

JSON File Exported

PROGRAM 11:

```
In [18]: import pandas as pd
import matplotlib.pyplot as plt
df = pd.read_csv("diabetes.csv")
df["Outcome"].value_counts().plot(kind="bar", color=["skyblue", "orange"])
plt.title("Distribution of Diabetes Outcome")
plt.xlabel("Outcome (0 = No Diabetes, 1 = Diabetes)")
plt.ylabel("Number of Patients")
plt.show()
```

